

TB clock and reset

Every RTL testbench needs a clock and a reset before stimulus runs

Starter classic assert times two

- Starter preset: classic assert times two, sync release
- Free-running clk toggles every half-cycle
- Rst_n stays low through two posedges
- Cursor starts at h4, the release point
- Click Analyze and the flags show assert_posedges equals two
- The TB sketch shows repeat two at posedge clk, then rst_n equals one

Browser lab

Digital Design and Verification Platform

HomeTools

Home / Tools / TB clock + reset patterns

INTERACTIVE TOOL

TB clock + reset patterns

Sketch TB **clock generation** and **reset assert / deassert** on a short timeline — how many cycles of `rst_n=0`, then release relative to `posedge clk`. Starter: hold reset two cycles, then release.

Starter example: free-running `clk`; `rst_n` low for two posedges, then release (classic sync-style).

Load starter example

Challenges 0 / 22 cleared

Quiz: clock: A typical TB clock is generated with...

☐ a forever toggle (or equivalent) in the testbench

☐ only the DUT's clock-gating cell

☐ \$finish inside the clock block

☐ randomize() on clk each ns

Show hint

Check

Next

Idle

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

21

22

Core ideas

forever clk

TB clock is a free-running toggle, not a DUT output.

Assert

Drive `rst_n=0` for N clock cycles at bring-up.

Sync release

Deassert near a clock edge so flops see a clean sample.

Async release

Change reset between edges — still legal; know the hazard.

TB clock reset starter

learn_uart — tb clock reset

Real RTL/TB practice

- In Track A, write a minimal initial block
- State why sync deassert is preferred for flops clocked on posedge clk
- Optional: peek at a UART TB in the legacy combined materials and name where reset ends
- This lab is literacy, not a full simulator or metastability proof

Pitfalls to watch

- Do not confuse TB clock generation with the UART baud divider, they serve different roles
- Assert means `rst_n` equals zero, not one
- Counting reset cycles means posedges while `rst_n` is low, not arbitrary half-cycles
- Releasing reset mid-cycle can work but is not the same as sync release at the edge
- Hold forever means the DUT never leaves reset, useful to see the wave, not a passing test
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser
- On paper, sketch `clk` and `rst_n` for two cycles in reset and one cycle out
- When you are ready, take the short quiz, then continue to waves